

STEREOCHEMISTRY

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Ques What is STEREOCHEMISTRY? The stereochemistry is the study of three dimensional configuration of the atoms that make up molecules and the ways in which the arrangement affects the physical and chemical properties of a molecule.
i.e. different structural position/relative position of atoms or groups are called stereoisomers.

Isomerism: Different compounds having the same molecular formula are called isomers. This phenomenon is known as isomerism.

Isomers: Those compound which have same molecular formula but different structure, or different name is known as isomers.

Isomerism: There are two main type of isomerism

(i) Structural Isomerism.

(ii) Stereoisomerism / space isomerism.

(i) Structural Isomerism: Structural isomerism have different structure formulae i.e. different arrangement of atoms within the molecule.

→ (A) Chain / or Nuclear isomerism

→ (B) Position isomerism

→ (C) Functional isomerism

→ (D) Metamerism

→ (E) Tautomerism

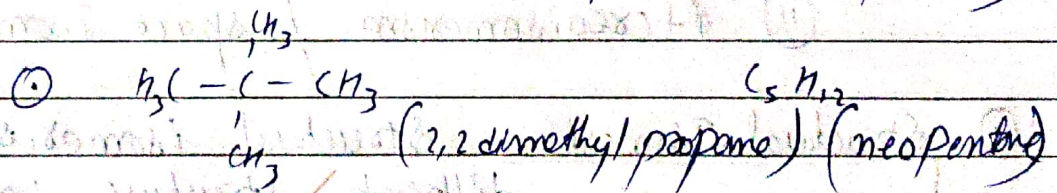
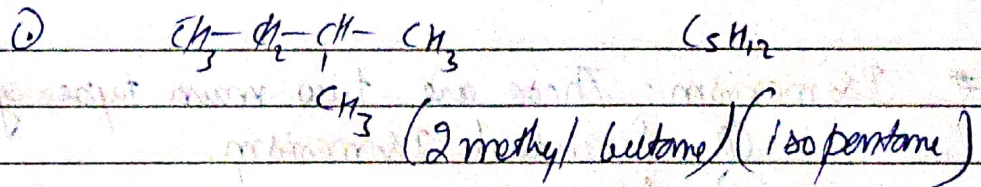
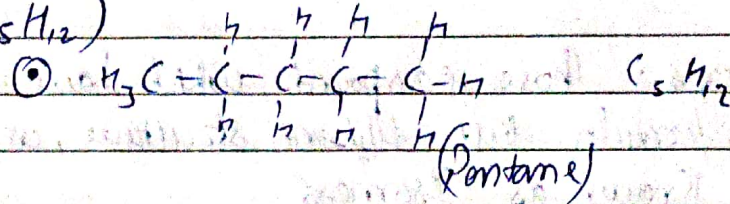
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(ii) Stereo isomerism: Stereoisomers have the same structure formula but different configuration i.e. different arrangement of atoms or group in space.

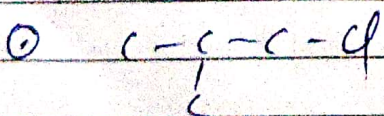
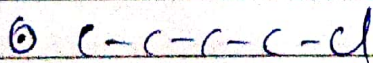
- (E) Geometrical isomerism
- (G) Optical isomerism
- (H) conformational isomerism.

(A) Chain Isomerism: Those compound which have the same molecular formula but different nature of carbon chain.

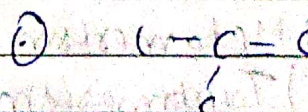
example ① Pentane (C_5H_{12})



④ Butyl chloride (C_4H_9Cl)

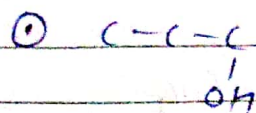
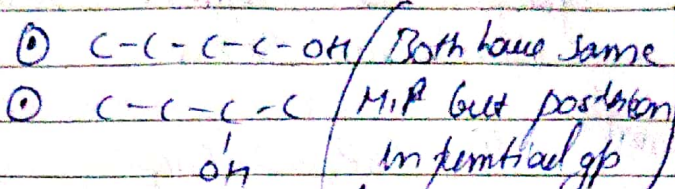
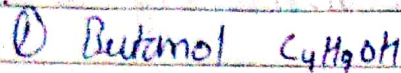


⑤ Butylene (C_4H_8) it have two chain isomers

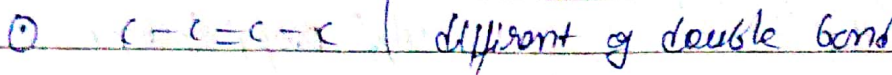
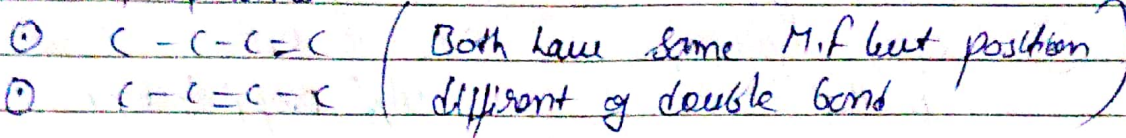


(B) Position Isomerism: Compounds which have same M.F but different in position occupied by an atom or functional group in a carbon chain.

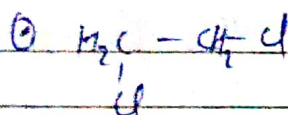
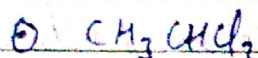
example



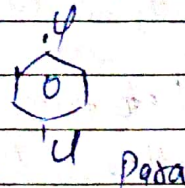
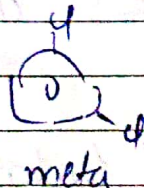
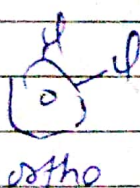
⑪ Butene exist in the form of two position isomers as under:



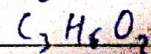
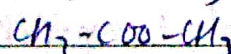
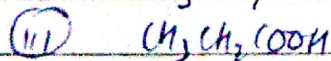
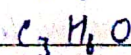
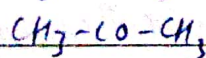
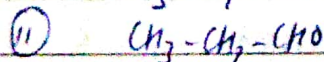
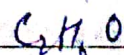
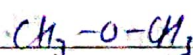
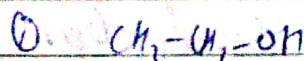
(11) 1,1 dichloro ethane $\text{C}_2\text{H}_4\text{Cl}_2$ and 1,2 dichloro ethane are position isomers.



(10) $\text{O}-\overset{\text{d}}{\text{m}}-\text{p}$ -Dichlorobenzene also show position isomerism.



(C) Functional Isomerism: Compound having same molecular formula but different functional groups.



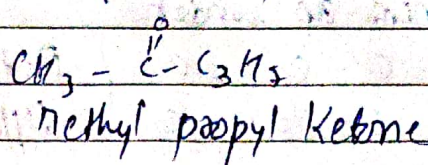
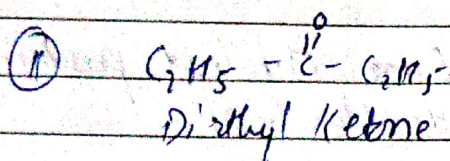
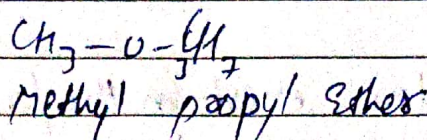
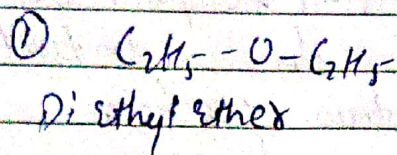
Methyl cyanide

Methyl isocyanide

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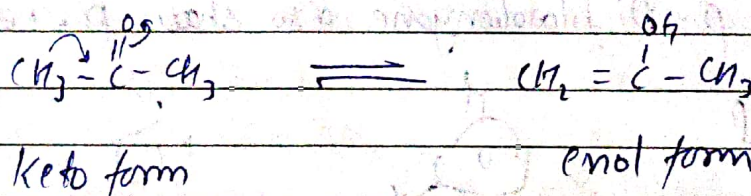
(D) Metamerism: Compound having same Molecular Formula and even the same functional group, but the two alkyl groups attached to the same functional group are different. These isomers belong to same homologous series.

example



(E) Tautomerism: It is a special case of functional isomerism and arises due to 1,3 migration of a proton from a carbon atom to another (oxygen/nitrogen) with the rearrangement of carbon-carbon single and double bond called tautomerism and this phenomenon is called tautomerism.

example



this is called Keto-enol tautomerism

① STEREOISOMERISM: The isomerism which arises due to difference in the arrangement of atoms in space is called stereoisomerism.

The isomers which have same Molecular formula but different configuration are called stereoisomers.

3 types

(A) Optical isomerism

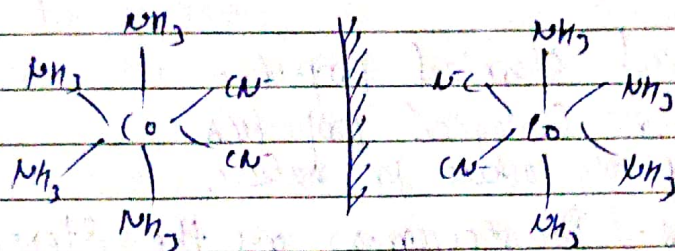
(B) Geometrical isomerism

(C) Conformational isomerism

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(A) OPTICAL ISOMERISM: It has been observed that certain compounds resemble one another in their chemical properties as well as in most of their physical properties but they differ in their behaviour towards the action of plane polarised light. Such compounds are called optically active and this phenomenon is called optical activity.

- * Compounds which rotate the plane polarised light towards the right are called dextro-rotatory and represented as (d) or (+).
 - * Compound which rotate Left called as Levo-rotatory represented as (l) or (-).
- These isomers are said to show optical isomerism.



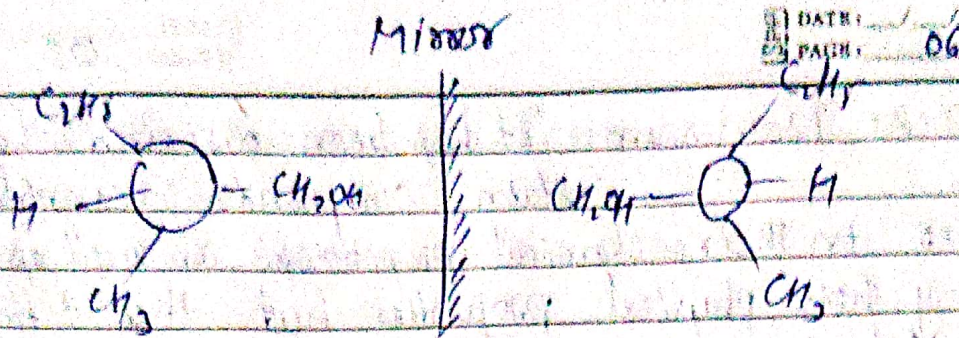
optically active

non superimposable

Type of Optical Isomerism:

- * Enantiomerism
- * Diastereomerism

- * Enantiomerism: Such isomers whose molecular structure are non-superimposable, mirror images of each other and which rotate the plane polarised light equally but in opposite direction are known as enantiomers. This phenomenon is known as enantiomerism.



Enantiomeric forms of 2-methyl-1-butanol

Essential Condition for Enantiomerism: A molecule is termed as chiral if it has no plane of symmetry and is, therefore, non-superimposable on its mirror image.

Characteristics of Enantiomers: (i) They have identical physical properties such as melting point, boiling point, densities, solubility.

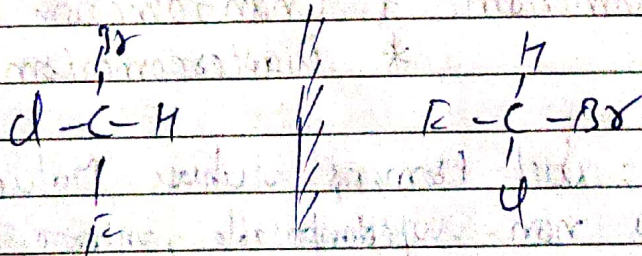
(ii) They have identical chemical properties.

(iii) They have different biological properties.

(iv) They always optically active in nature.

* Diastereomers: Diastereomers are the stereoisomers that are non-superimposable and are not mirror images.

example



* Different molecular shape.

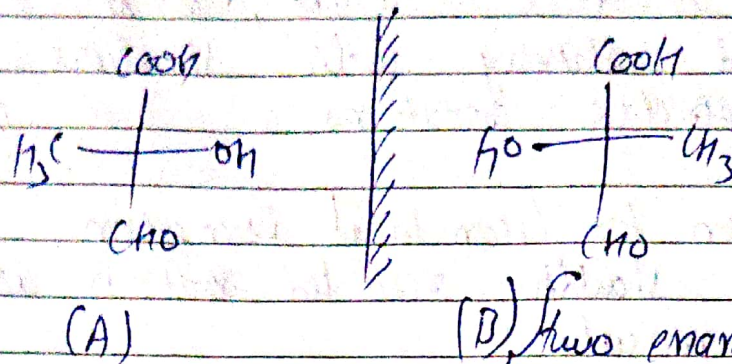
* have different physical properties.

* have different chemical properties.

* They are optically active or optically inactive in nature.

RACEMIC MIXTURE: Racemic mixture contains the equal or equimolar amount of two enantiomers pair is called racemic mixture.

Example

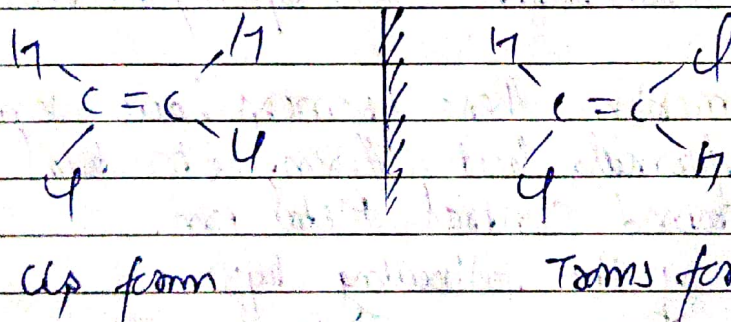


(B) Two enantiomers pair
non-superimposable

* So we take x mole from A also x mole from B and mix it well. and the mixture we get is known as racemic mixture.

* we get racemic by adding equal quantity of dextro (+) and levo (-)

(B) GEOMETRICAL ISOMERISM: The isomerism which arises due to restricted rotation about double bond in molecules [OR] about single bond in cyclic compounds is known as Geometrical or cis-trans isomerism.



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① Geometrical Isomerism shown by variety of compounds such as:-

→ Compound containing double bond



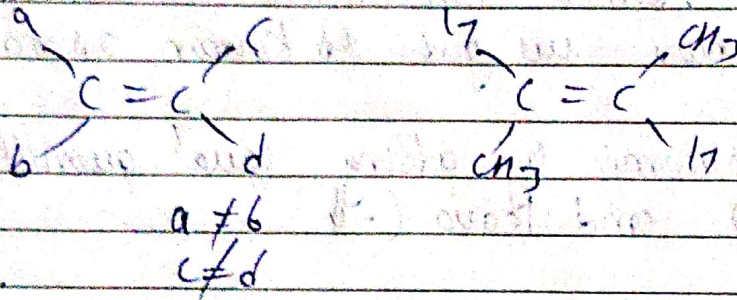
→ Compound containing cyclic structure

→ Restricted rotations

② Condition for Geometrical Isomerism:

① There should be restricted rotation about a Bond in a molecule

② Both substituents on each carbon about which rotation is restricted should be different



Important Question

Discuss the different type of Isomerism possible in transition metal compounds.

There are mainly two type of isomerism possible in transition Metal complex.

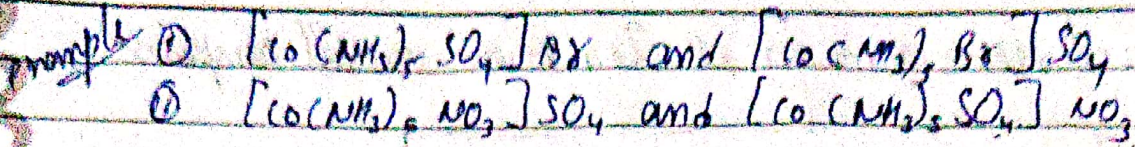
(A) Structural Isomerism (B) Space/ Stereo Isomerism.

(A) Structural Isomerism. These isomers have same Molecular formula but different structural arrangements of atom around central Metal ion.

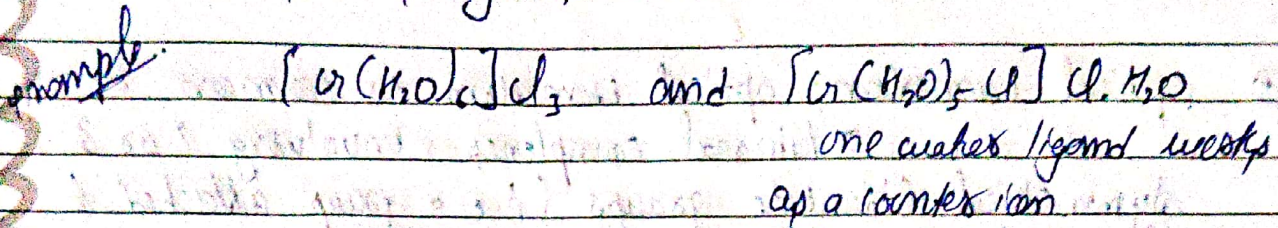
It is mainly of the following types.

① Ionisation Isomerism: Those isomers which have same Molecular formula but different anions and cations in solution to different counter ion.

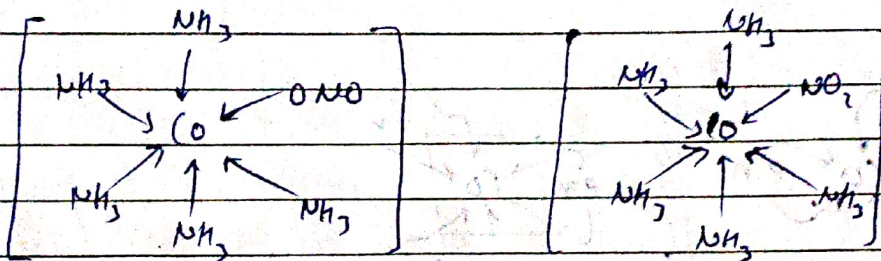
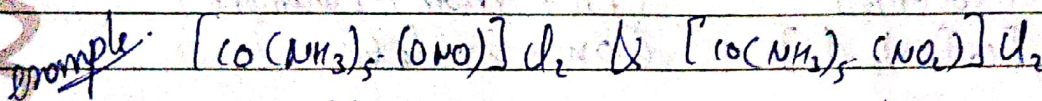
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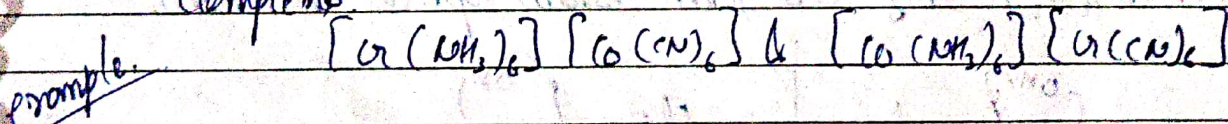
⑪ Hydrate Isomerism: / Solvate Isomerism, isomers which have same molecular formula but differ in term of diff. H_2O molecules as ligands.



⑫ Linkage Isomerism: Isomers which have same M.F. but isomerism occurs in complex compound which contain ambidentate ligands.



⑬ Co-ordination Isomerism: Compound having Molecular Formula same but different in terms of ligands in coordination complex. It contain two complexes.



(B) Space / Stereoisomerism in Transition metal compound

① Geometrical Isomerism

- ↳ Cis form
- ↳ Trans form

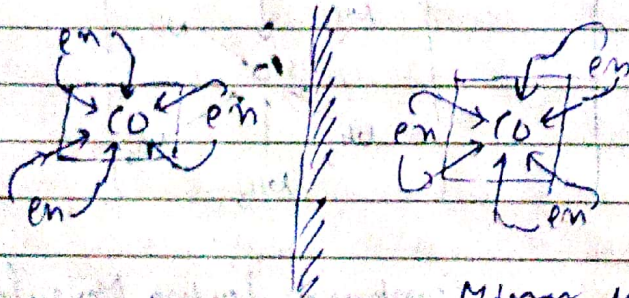
② Optical Isomerism

- ↳ Dextro (+)
- ↳ Levo (-)

③ Optical Isomerism: optical isomerism is common in octahedral complexes involving 2 or 3 symmetrical bidentate groups (i.e. a group attached to the central atom by two co-ordinate bonds).
like eg. $[M(CAA)_2X_2]$ and $[M(CAA)_3]$
where AA = bidentate ligand
M = Central Metal.

Example (A) $[Co(en)_3]$ (en = ethane-1,2-diamine)
(B) $[Co(ox)_2Cl_2]$ (ox = oxalato)
cis

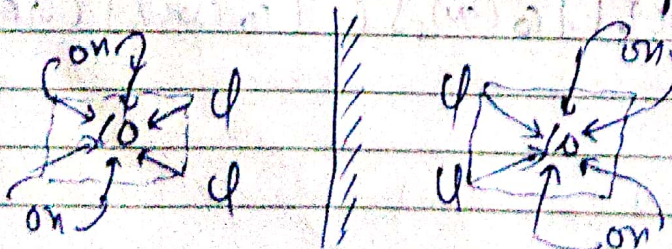
(A)



Mirror image non-superimposable
so they are optically active.

Also called enantiomers.

(B) $[Co(ox)_2Cl_2]$ only cis form show optical isomerism.

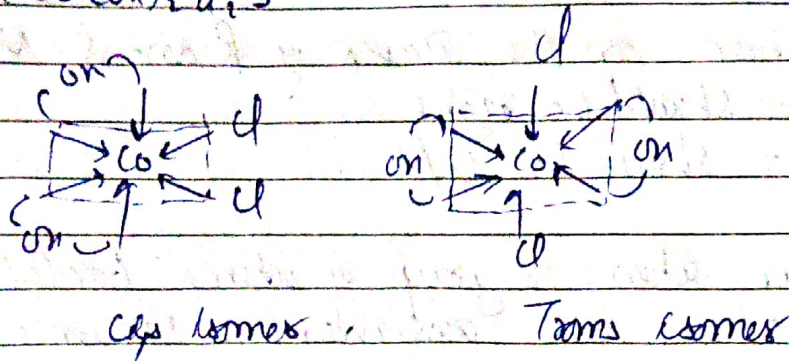


non-superimposable
so optically active
on cis form of the
compound.

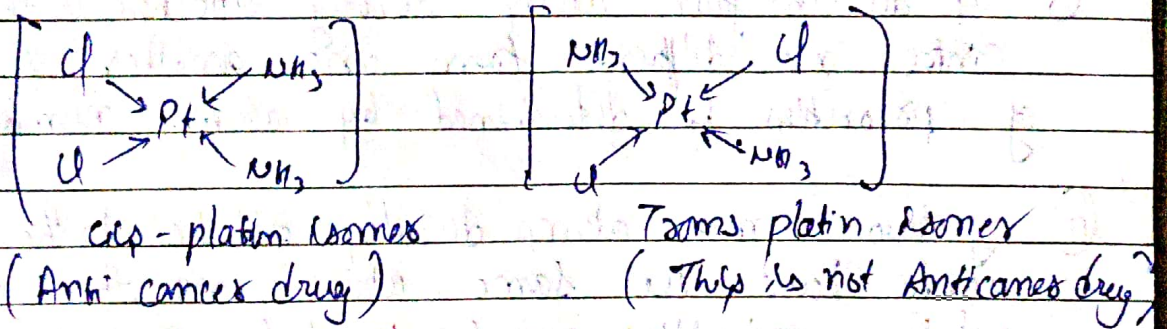
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- ① Geometrical isomerism: Geometrical isomerism differs in the spatial distribution of atoms or groups around central atom. Those complexes in which the two same ligands occupy adjacent position to each other are called cis-isomer while ligands occupying position opposite to each other are called trans-isomer.

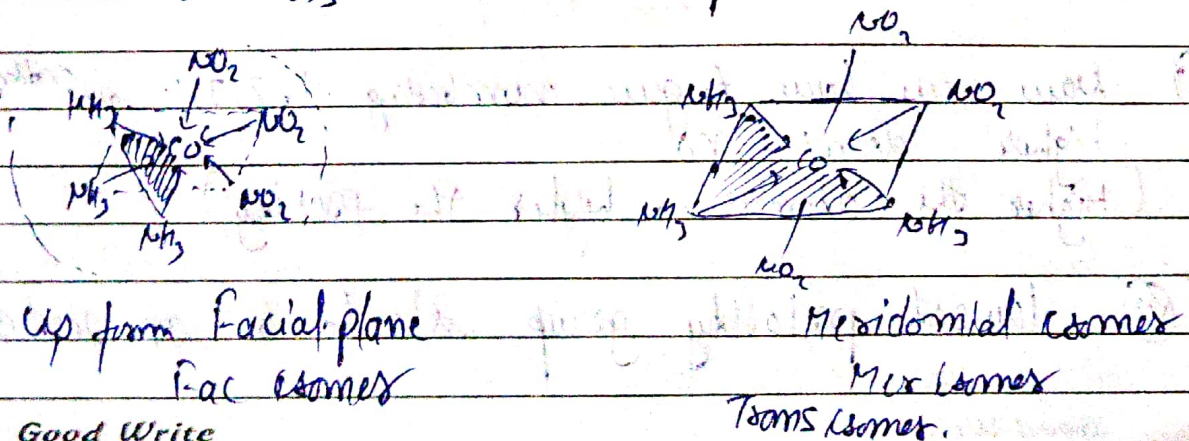
Example ① $[Co(NH_3)_2Cl_2]$



② $[Pt(NH_3)_2Cl_2]$ tetrahedral complex

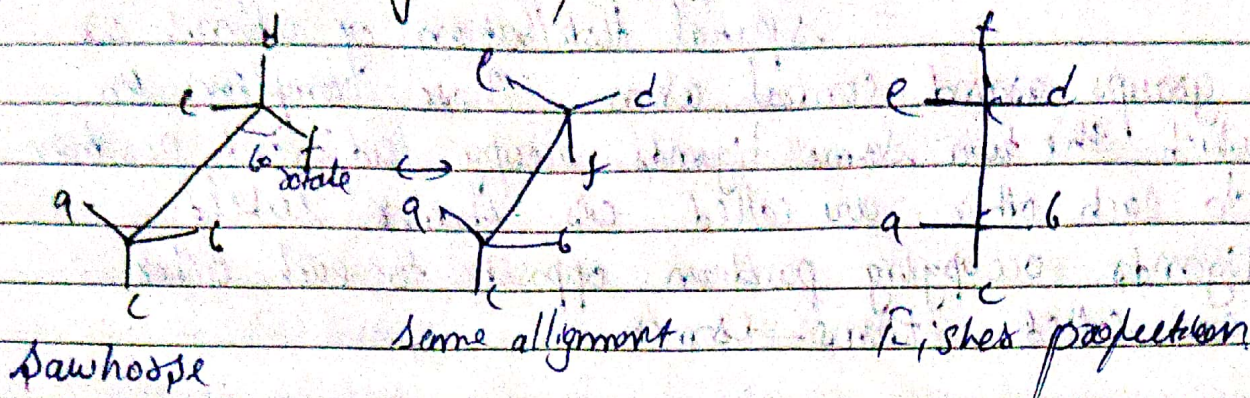


③ $[Co(NH_3)(NO_2)_5]$ octahedral complex



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Conversion of Perspective Sawhorse to Fisher.



Configuration on the Basis of R and S NOTATIONS

R = should be right

S = should be left

The four atoms or group of atoms bonded to the chiral centre are assigned a sequence of priorities.

The sequence rules are as follows:

(i) If all the four atoms directly attached to the chiral centre are different from one another, sequence of priorities is determined by atomic numbers.

(ii) If two or more atoms directly attached to the chiral centre have the same atomic no., then the relative priorities may be fixed by comparing the next atoms in the groups.

(iii) Now we have to give numbering 1, 2, 3, 4 according to higher atomic no.
(Higher the atomic no. higher the priority)

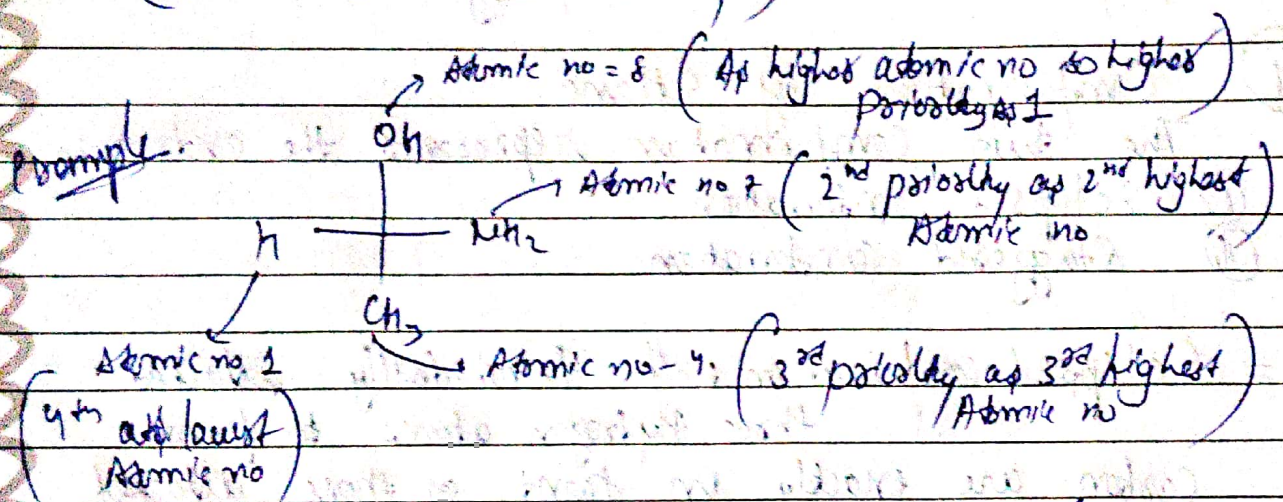
imp. (iv) lowest priority group should be on vertical line

⑤ Now we have to make parallel transition from horizontal to vertical

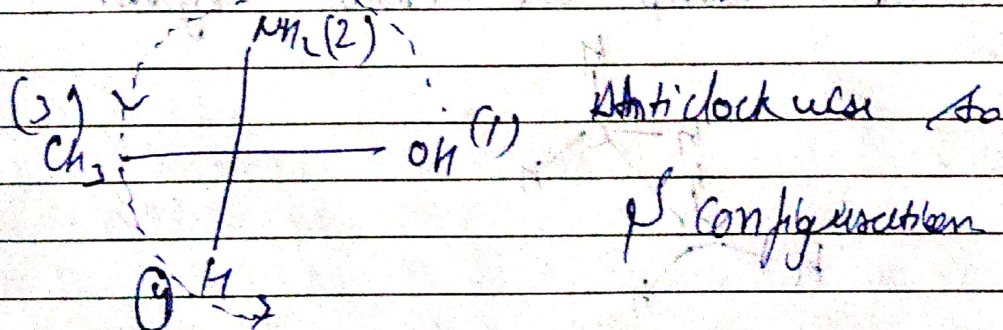
By shifting the groups we get the new structure with same configuration.

⑥ Now we have to travel with priority groups. As if in doing so, the eye travels in clockwise direction, the configuration is specified as R (Latin word: rectus = right).

If on the other hand, the eye travels in anticlockwise direction, the configuration is specified as S (Latin word: sinister = left).



As lower priority group should be on vertical so



Conformation of Alkanes

- Rotation around C-C single bond. Single covalent bond (σ -Bond) present. B/w two carbon atoms is formed by the overlap of their sp^3 hybrid orbitals along the inter nuclear axis.
- Due to the axial symmetry of the molecule exists, rotation around the C-C bond is almost free.
- As a result of this rotation, alkanes can have different spatial arrangements i.e. different relative arrangement of their atoms in space. This is called the conformation of alkanes.

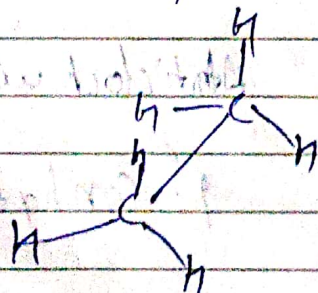
① Conformation of Ethane.

The two conformation represents the extremes.

① Eclipsed conformation

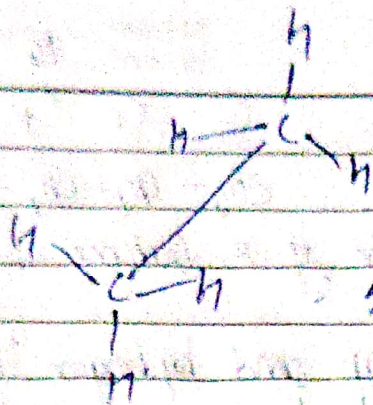
② Staggered conformation.

= Eclipsed conformation: the tetrahedrally attached three hydrogen atoms to front carbon are exactly in front of those attached to the back carbon i.e. hydrogen atoms of both the carbon atoms are crowded together.



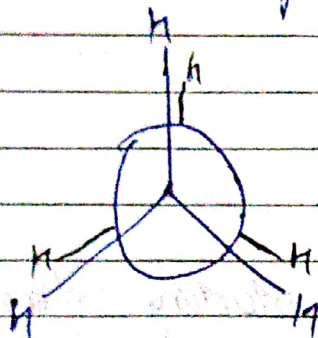
= Staggered conformation: the tetrahedrally attached hydrogen atoms to two carbon atoms are as far apart as possible.

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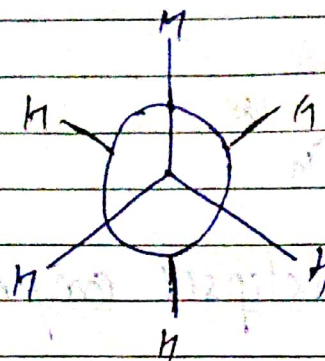


staggered conformation of Ethane.

Newman's style representation the conformation of molecules
The two conformations of Ethane

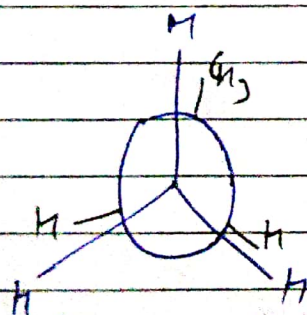


Eclipsed

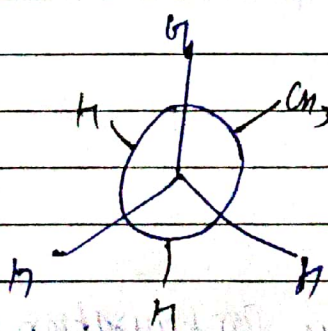


staggered.

Conformation of Propane. $\text{CH}_3-\text{CH}_2-\text{CH}_3$



Eclipsed.

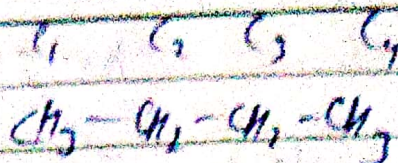


staggered

Very imp

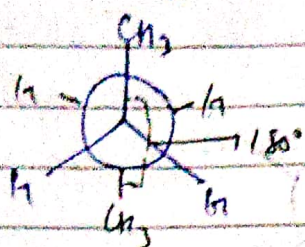
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Conformation of n-Butane

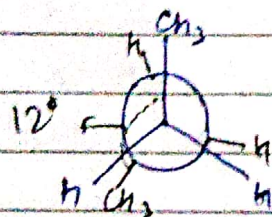


The various possible conformation of n-butane

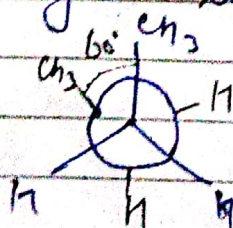
- (i) Anti-Conformation - Methyl group and hydrogen linked staggered to two carbon atoms C_2 and C_3 are at a maximum distance apart. It is assumed that the angle of rotation around the $C_2 - C_3$ bond is zero. For this conformation



- (ii) Partially eclipsed conformation: Starting from the anti-conformation, if we rotate the $C_2 - C_3$ single bond through 60° , the methyl group attached to one carbon atom is at the back of hydrogen atom attached to the other carbon atom.



- (iii) Gauche / Skew Conformation: On rotating $C_2 - C_3$ single bond by another 60° the various bonds are farthest apart but the methyl groups make an angle of 60° with respect to each other.



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- (16) Fully eclipsed conformation: Rotating the C_2-C_3 bond by another 60° gives rise to fully eclipsed, the methyl group and hydrogen atoms attached to two carbon atoms completely eclipse each other.

